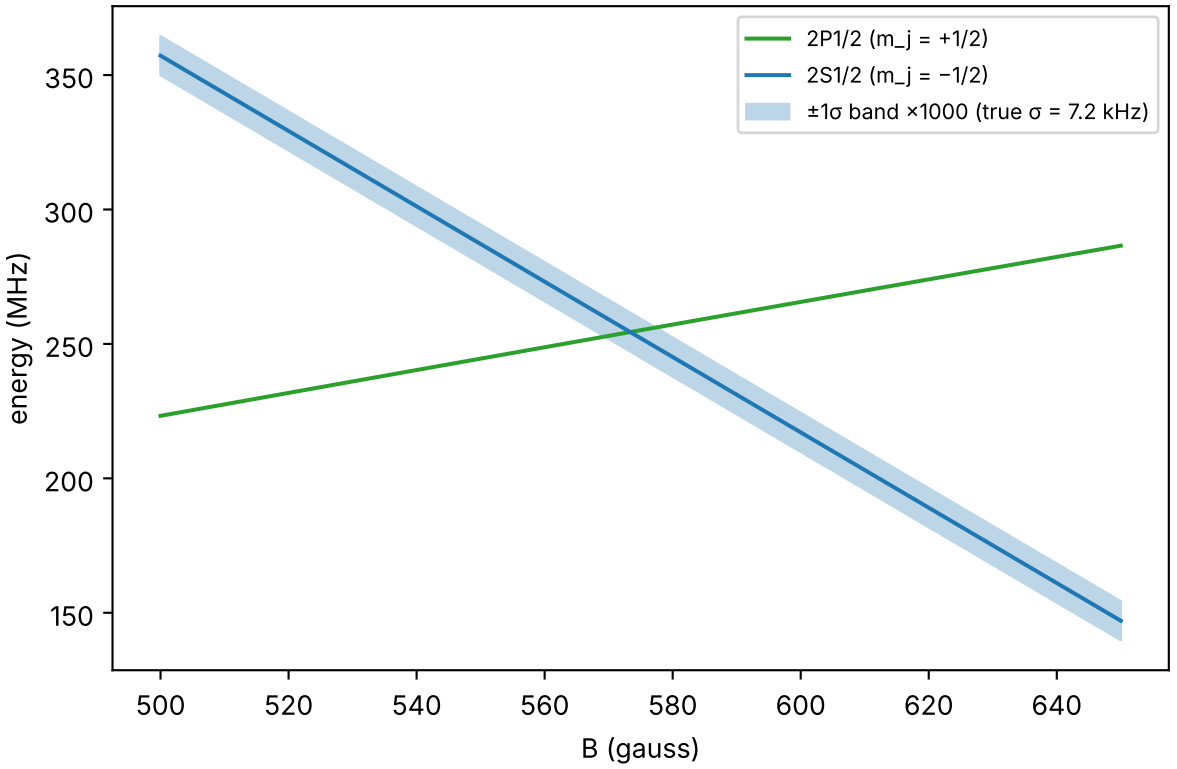
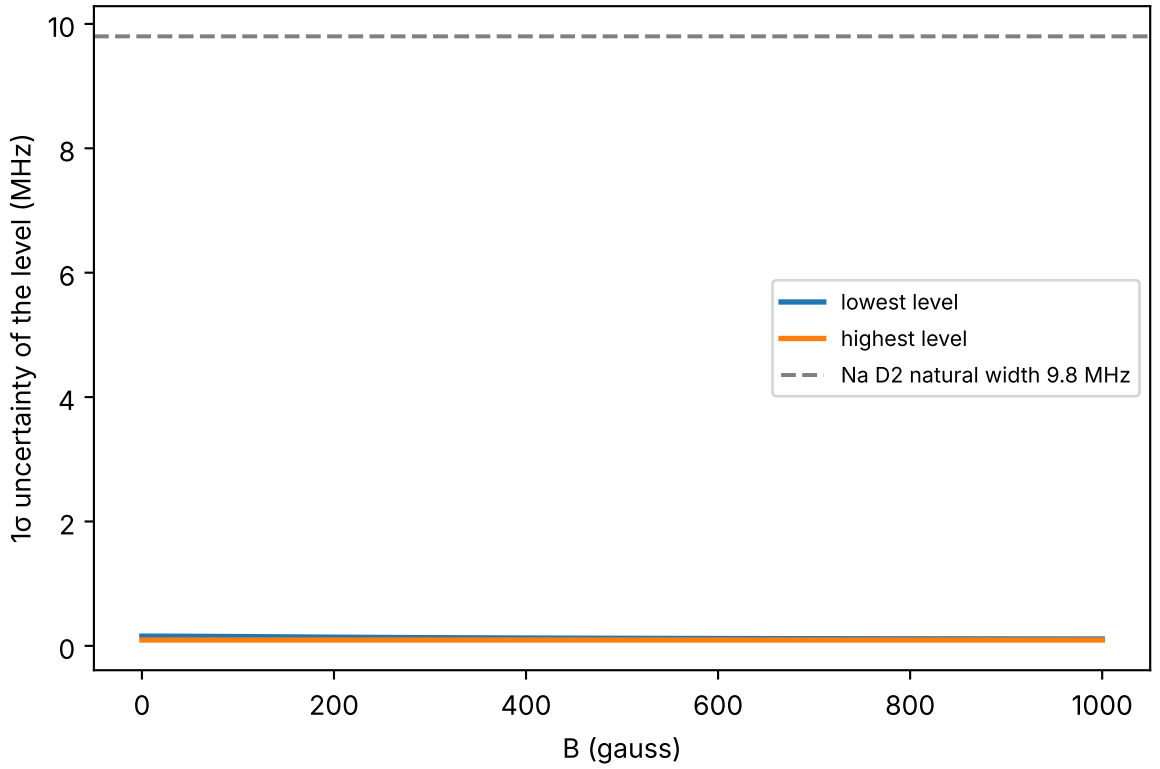

H crossing: Lamb-shift uncertainty 0.009 MHz moves the 2S level
by far less than the 45 to 110 MHz line widths of Westig et al.
2P1/2 (m_j = +1/2)
2S1/2 (m_j = −1/2)
±1σ band ×1000 (true σ = 7.2 kHz)
energy (MHz)
350
300
250
200
150
500
520
540
560
580
600
620
640
B (gauss)


Sodium 3P1/2 hyperfine levels: uncertainty from A(3P1/2)
is tenths of a MHz, below the natural width
lowest level
highest level
Na D2 natural width 9.8 MHz
1σ uncertainty of the level (MHz)
10
8
6
4
2
0
0
200
400
600
800
1000
B (gauss)